Notes: Candidate Energy Ansatz for 2D Maxwell–PML

Working notes, not a proof

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1 Purpose

This note lists candidate structures to try. It should not be cited as a proof. The main source problem is `energy_functional_problem.tex`.

2 Naive Energy and Its Defect

For the displayed model, first take the non-fractional case $\alpha = 1$:

$$P + \tau \partial_t P = \chi E, \qquad \chi = \epsilon_0(\epsilon_s - \epsilon_\infty).$$

The naive electromagnetic energy is

$$\mathcal{E}_{\mathrm{em}} = \frac{1}{2}\int_\Omega \epsilon_0\epsilon_\infty E^2 + \mu H_x^2 + \mu H_y^2 \,\mathrm{d}x\,\mathrm{d}y.$$

Under zero boundary flux, its derivative contains

$$\begin{aligned} &-\epsilon_\infty(\sigma_x+\sigma_y)E^2 - E\partial_t P - \frac{\sigma_x+\sigma_y}{\epsilon_0}EP + E(P_{z,1}+P_{z,2}) \\ &-\frac{\mu}{\epsilon_0}(\sigma_y-\sigma_x)H_x^2 - \frac{\mu}{\epsilon_0}(\sigma_x-\sigma_y)H_y^2 + H_xQ_x + H_yQ_y. \end{aligned}$$

The term $-E\partial_t P$ is favorable when $\alpha = 1$, because

$$E\partial_t P = \frac{1}{2\chi}\frac{\mathrm{d}}{\mathrm{d}t}\|P\|^2 + \frac{\tau}{\chi}\|\partial_t P\|^2.$$

Thus a natural first correction is

$$\mathcal{E}_{\mathrm{em+P}} = \mathcal{E}_{\mathrm{em}} + \frac{1}{2\chi}\|P\|^2.$$

The remaining electric PML terms involving EP, $EP_{z,1}$, $EP_{z,2}$, and the magnetic sign-indefinite terms still need auxiliary-variable structure.

3 Magnetic Shift

Try

$$M_x = Q_x - \frac{\mu}{\varepsilon_0}(\sigma_y - \sigma_x)H_x, \qquad M_y = Q_y - \frac{\mu}{\varepsilon_0}(\sigma_x - \sigma_y)H_y.$$

The model then gives

$$\partial_t M_x = \frac{\sigma_y - \sigma_x}{\varepsilon_0}\partial_y E - \frac{\sigma_y}{\varepsilon_0}M_x, \qquad \partial_t M_y = -\frac{\sigma_x - \sigma_y}{\varepsilon_0}\partial_x E - \frac{\sigma_x}{\varepsilon_0}M_y.$$

These identities suggest trying an energy in $E, H_x, H_y, P, P_{z,1}, P_{z,2}, M_x, M_y$. A possible family is

$$\begin{aligned}\mathcal{E}_A = \mathcal{E}_{\mathrm{em+P}} &+ \frac{1}{2}\int_\Omega \alpha_1 P_{z,1}^2 + \alpha_2 P_{z,2}^2 + \alpha_x M_x^2 + \alpha_y M_y^2 \,\mathrm{d}x\,\mathrm{d}y \\ &+ \int_\Omega \beta_1 E P_{z,1} + \beta_2 E P_{z,2} + \beta_3 P P_{z,2} \\ &+ \beta_x H_x M_x + \beta_y H_y M_y \\ &+ \gamma_x E\,\partial_y^{-1} M_x + \gamma_y E\,\partial_x^{-1} M_y \,\mathrm{d}x\,\mathrm{d}y,\end{aligned}$$

where the last two nonlocal terms are only a brainstorming device. Prefer local terms if possible.

4 Finite-Dimensional Lyapunov Test

For constant σ_x, σ_y, Fourier transform in space. For each $k = (k_x, k_y)$, write

$$\frac{\mathrm{d}}{\mathrm{d}t}\widehat{U} = A(k, \sigma_x, \sigma_y)\widehat{U}.$$

Then search for $G = G(k, \sigma_x, \sigma_y)$ with

$$G = G^* > 0, \qquad GA + A^*G \le 0.$$

If G can be chosen independent of k, it is a strong hint for a local L^2 energy. If G depends on k, translate it into derivative or pseudo-differential energy only with care.

5 Cole–Cole Memory

For $0 < \alpha < 1$, expect a positive memory term of the schematic form

$$\mathcal{E}_{CC} \sim c_P\|P\|^2 + \tau_0^\alpha \int_0^\infty w(\lambda)\|\phi(\lambda)\|^2 \,\mathrm{d}\lambda,$$

but the constants c_P and weight $w(\lambda)$ must be derived. The derivation should use

$$P + \tau^\alpha \partial_t^\alpha P = \epsilon_0(\epsilon_s - \epsilon_\infty)E,$$

or an equivalent diffusive representation. This should be treated only after the $\alpha = 1$ case is understood.